TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Driven Smart Insole for Advanced Gait Analysis and Clinical Rehabilitation Applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio/Medical		Bio/Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Therapeutic Devices & Diagnostic Devices	Intelligent Interpretation Systems
Concept of Proposed Technology	● **DNN-based smart-insole gait analysis** By integrating gait sensor data collected from smart insoles (e.g. plantar pressure and IMU signals) with clinical information, the proposed technology aims to develop a **'DNN-based Integrated Gait Analysis and Prognosis Prediction Solution'**. To ensure real-world usability, we tailor model architectures and data-encoding strategies to improve generalization. The solution is designed to: ① (Objective Evaluation) Objectively quantify gait impairment and abnormal patterns to support clinical assessment beyond subjective qualitative evaluation. ② (Prognostic Prediction) Predict functional outcomes after surgery or treatment in patients with lower-limb musculoskeletal conditions and classify patients as responders or non-responders. ③ (Personalized Intervention) Based on the evaluation and prediction results, the system supports clinical decision-making for individualized rehabilitation strategies.			

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Rationale for Joint R&D Project

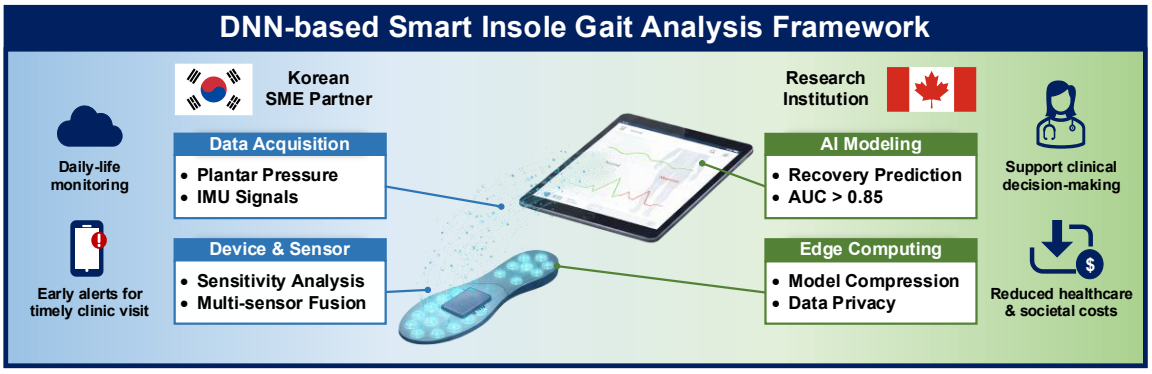

Fig. 1. Conceptual overview of the proposed DNN-based smart-insole gait analysis framework

1. Necessity of the Proposed Technology

① (Rising Burden): Chronic diseases and gait impairment are increasing, raising healthcare and caregiving costs

② (Need for paradigm shift): Healthcare must move from post-event treatment to prevention, early risk prediction, and continuous management.

③ (Current gap): Preventive care depends on self-initiated, limited clinic visits, so conditions are frequently detected late [1]

2. Necessity for Collaborative Research

This project aims to establish an integrated framework that links smart insole-based gait data with clinical information to enable DNN-based abnormal gait analysis, early risk prediction of geriatric conditions, and rehabilitation support. The proposed technology includes the following tasks:

① (Data Collection) Develop preprocessing protocols utilizing high-precision sensors capable of measuring foot pressure at 15+ key joint locations—an essential element for gait analysis.

② (Algorithm/Modeling) Develop DNN models that integrate smart-insole data and clinical information for abnormal gait classification, risk score estimation.

③ (System/Edge) Design a low-latency inference framework by embedding gait analysis algorithms into a cross-disciplinary SW platform capable of integrating diverse clinical fields.

④ (Clinical Integration) Integrate model outputs within clinical workflow (e.g., trends, alerts).

A single organization cannot simultaneously provide the specialized expertise and validation framework. Clinical deployment of medical AI requires clinical problem definition, IRB-approved data acquisition, clinical validity evaluation and field feedback, while also addressing operational constraints such as security and latency for real-world deployment. The project team consists of **(industry) Korean SME Partner** (device development, productization, service integration, and coordination for clinical requirements and validation), and **Research Institution** (model development, model compression, and edge optimization), working in a complementary manner. [2]

3. Rationale for Collaboration

① (Interdisciplinary expertise) The project spans wearables, bio signal processing, time-series AI, clinical research, and edge systems, requiring multidisciplinary expertise.

② (Cross-domain synergy) Joint development is needed for an end-to-end framework.

③ (Need for field validation) Field trials are required to verify robustness in diverse environments.

Key Words

● gait analysis, artificial intelligence, edge computing, healthcare, wearable

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (8)

R&D Objectives

Korean SME Partner

Upgrade commercially available smart insoles currently used in the leisure sports domain into commercialization-level devices suitable for clinical use in rehabilitation medicine.

- Development of commercialization services for an AI solution that supports gait disorder diagnosis and prognosis prediction, as well as a gait rehabilitation intervention solution, based on an existing commercial smart insole product
- Development of a new high-performance smart insole product to enhance measurement precision to a level suitable for clinical research and diagnostic testing in the medical field
- Development of technologies that utilize smart insoles and artificial intelligence to assess the degree of gait function deterioration in patients with lower-limb joint disorders, predict prognosis, develop patient-specific intervention strategies, and evaluate the feasibility of early clinical applications.

Research Institute

Development of advanced artificial intelligence solutions for gait analysis, and development of a DNN-based prognostic prediction model for patients with lower-limb joint disorders.

- A multimodal DNN-based prediction technology that integrates gait sensor data from smart insoles with clinical information to predict postoperative functional outcomes (responder vs. non-responder) in patients with lower-limb joint disorders.
- Model architecture and data encoding technologies optimized for multichannel, time-series sensor data, enabling robust feature extraction and reliable performance in real-world clinical environments.
- Model interpretability and sensor contribution analysis techniques that quantify the impact of individual sensor signals on prediction outcomes, enhancing clinical trust and usability.
- Data-driven hardware optimization supports technology, where insights from model analysis are utilized to inform smart insole hardware design and sensor performance enhancement.

R&D Target Specifications

Korean SME Partner

Quantitative Specifications of the High-Performance Smart Insole Product

- A 6-axis sensor and pressure sensors for measuring plantar pressure at up to 16 or more key foot regions
- Application of BLE 5.0 to ensure large-volume data handling and high-speed transmission
- Adoption of a rechargeable battery system considering usability and user acceptance
- Application of top-tier domestic durability testing standards conducted by accredited testing institutions
- Acquisition of domestic and international official certifications for product release (KC, FCC, CE, RoHS, etc.)

Research and clinical output targets

- Development of a prognostic prediction model for postoperative outcomes based on gait function in patients with lower-limb joint disorders (1 SCI-indexed international journal publication)
- Exploration of the clinical applicability of rehabilitation interventions (1 clinical research protocol and 1 IRB approval)

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R&D Plan

Research Institute

Degree of improvement over existing sensor-signal-based gait analysis AI models

- Development and validation of a DNN-based prognostic prediction model that integrates time-series gait sensor from smart insoles with clinical information to predict postoperative functional recovery in patients with lower-limb joint disorders.
- The proposed model aims to achieve an Area Under the ROC Curve (AUC) of at least 0.85 in predicting postoperative functional outcomes, as evaluated on test datasets.
- The model will provide quantitative interpretability metrics by analyzing the contribution of major sensor channels of plantar pressure and inertial acceleration signals.
- Based on the developed technologies and validated performance, the project targets the submission of at least three SCI-indexed international journal publications.

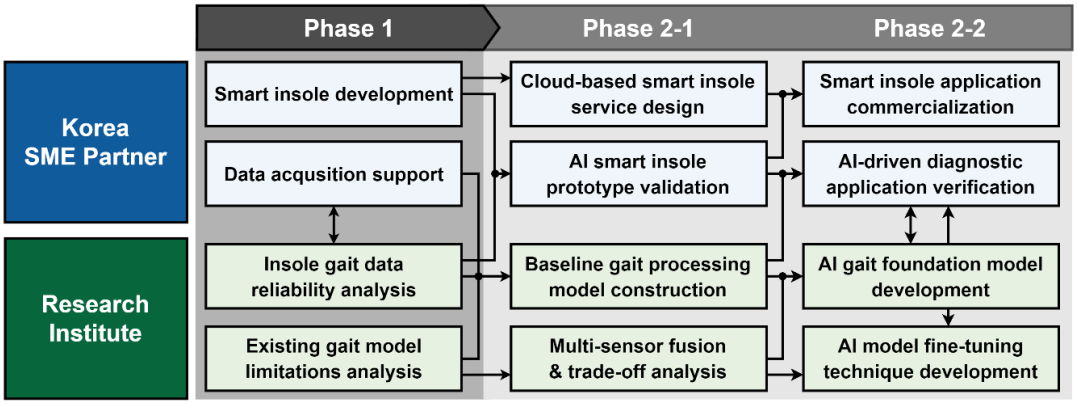

Fig. 2. Overall R&D Roadmap

Phase 1

Korean SME Partner

Development of gait analysis insoles through hospital-research institute collaboration, including environment setup and core HW/SW IP provision.

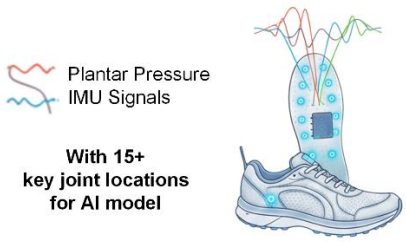

Fig. 3. Example of High-performance Smart Insole Product Design

- Development and supply of smart insole and integrated solutions for data acquisition to support AI-based research and development
- Integrating approaches to ensure a seamless bridge between physical sensor hardware and high-level AI analysis environments
- Provision of on-site technical support for data collection, as well as data preprocessing and refinement to facilitate AI model development
- Circuit design of high-performance smart insoles, fabrication of prototype samples, and validation of performance and reliability

- Adopting rigorous testing protocols to guarantee that all hardware components maintain precision and durability even in demanding physical environments.

Research Institute

Analyzing smart insole data reliability and identified limitations in existing AI/signal processing models for gait analysis

1. Fundamental Reliability Analysis of Smart Insole Data

- Identify and characterize critical data quality issues – specifically signal noise, sensor drift, high variability – to establish a ground truth for data integrity and define the boundaries of reliable measurement

2. Drawbacks for Conventional Signal Processing Approaches

- Structural limitations of traditional rule-based detection methods (e.g., fixed thresholding and windowing)

3. Existing Model Constraints

- "Black Box" opacity of current models obscures the biomechanical rationale behind predictions, preventing clinicians from validating diagnostic logic

Fig. 4. Limitation of conventional gait analysis approaches under smart insole data characteristics

Phase 2-1

Korean SME Partner

Enhance core components and firmware for gait-specific analysis while facilitating clinical deployment of Research Institute's AI models

1. Data Acquisition & Processing Support to Research Institute.

- Provide on-site technical support for secondary data collection and data processing to facilitate AI research.
- By implementing automated data labeling pipelines and rigorous cleaning protocols, ensure the high fidelity and integrity of datasets required for deep learning model training.
- Investigating whether gait indicators can serve as digital biomarkers representing structural transformations related to aging and disease, such as intramuscular fat infiltration.
- Establishing a high-fidelity mapping between spatiotemporal gait metrics—such as stride variability and gait speed—and musculoskeletal health indicators to ensure diagnostic accuracy.

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2. Cloud & Mobile AI API Infrastructure

- Design and construct cloud and mobile-based API systems to host and deploy AI models.
- The infrastructure will utilize a scalable microservices architecture, ensuring low-latency inference and secure data transmission between the smart insole sensors and the cloud database for preserving privacy.

3. Application Planning & Prototype Development

- Plan, design, and develop prototype concepts for AI applications aimed at successful commercialization.
- This process includes comprehensive user journey mapping and the creation of high-fidelity wireframes to validate the service logic and ensure a seamless experience for both clinical staff and elderly end-users.
- With feedback provided by severance, improve the clinical utility of the insole hardware and analysis solutions, also AI modeling feedback from research institute.

4. Advanced Smart Insole Engineering

- Advanced design of high-performance smart insole prototypes, including sample production and validation.
- Emphasis will be placed on ergonomic material selection for long-term comfort and precise sensor calibration to capture subtle changes in gait patterns, which are critical for early geriatric disease detection.

Test Item	Comparative Value	Verification Method
Pressure Sensor **High-load Impact** Durability	Global Max (Over 1 million cycles)	Accredited Testing Agency
Pressure Sensor **Bending** Durability	Global Max (Over 1 million cycles)	Accredited Testing Agency

Table. 1. Evaluation criteria for medical sensor/firmware improvements

Research Institute

Establish a baseline gait analysis model using clinical data and developed scenarios to maximize its utility in field applications

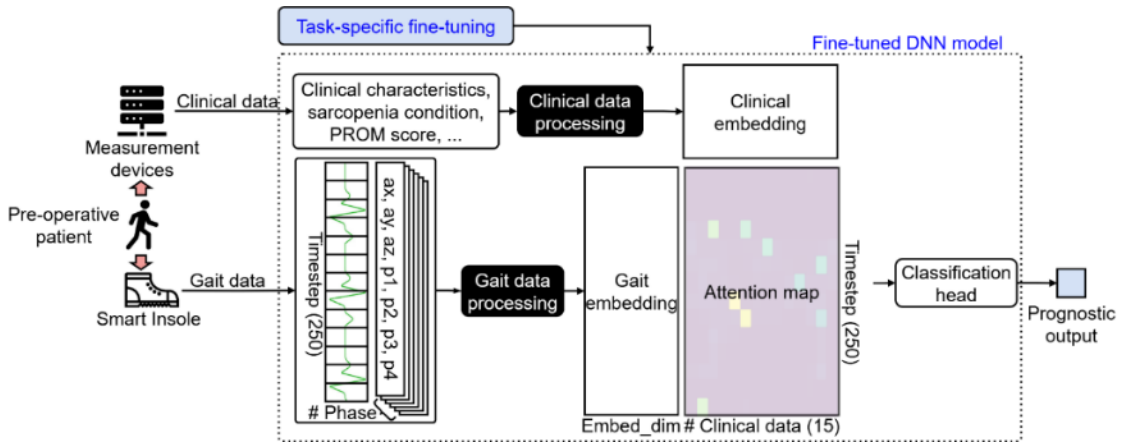

Fig. 5. Insole foundation DNN model and fine-tuning framework for variable gait tasks

1. Baseline Model Construction & Feedback Integration Infrastructure

- Establishment of a baseline model capable of addressing clinical challenges using standardized data.

2. Multi-Sensor Fusion & Trade-off Analysis

- Analysis of the trade-offs between performance gains from additional sensor data and the corresponding increase in model complexity.
- Employ optimal multi-sensor fusion strategies—such as feature-level or decision-level fusion—to maximize diagnostic accuracy while maintaining the low-latency requirements necessary for real-time processing.

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Phase 2-2

Korean SME Partner

Secure initial gait analysis prototypes for commercialization based on clinical validation and system compatibility

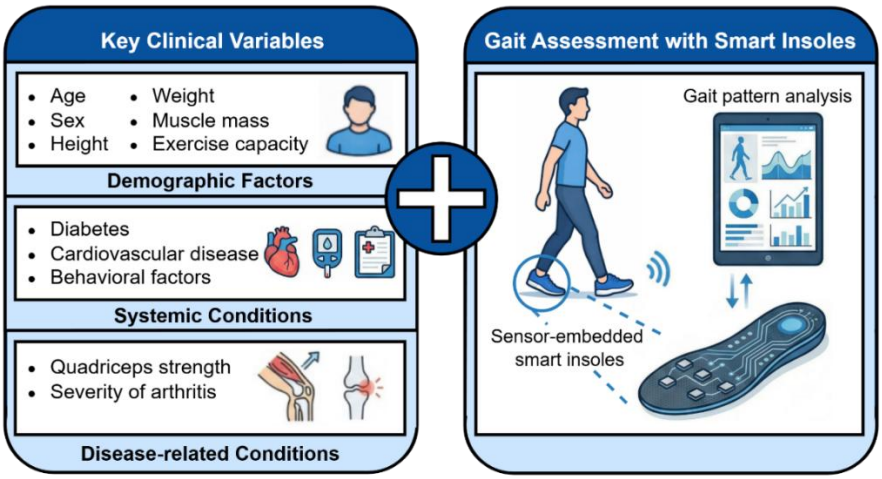

Fig. 6. Clinical problem definition and data acquisition framework

1. AI-Driven Diagnostic Application for Geriatric Diseases

- Advanced development through the prototyping of an application service based on AI-assisted diagnostic solutions for geriatric diseases, followed by validation through clinical field applications in severance.
- By optimizing the AI inference engine for real-time gait analysis and developing a user-centric interface (UI/UX), it can provide professional healthcare solutions for elderly patients.

2. High-Performance Smart Insole Prototype & Packaging

- Production and validation of high-performance smart insole prototypes, alongside the development of commercial-grade packaging.
- The development will focus on enhancing sensor durability and battery efficiency, ensuring the hardware can withstand long-term daily use while maintaining data accuracy.

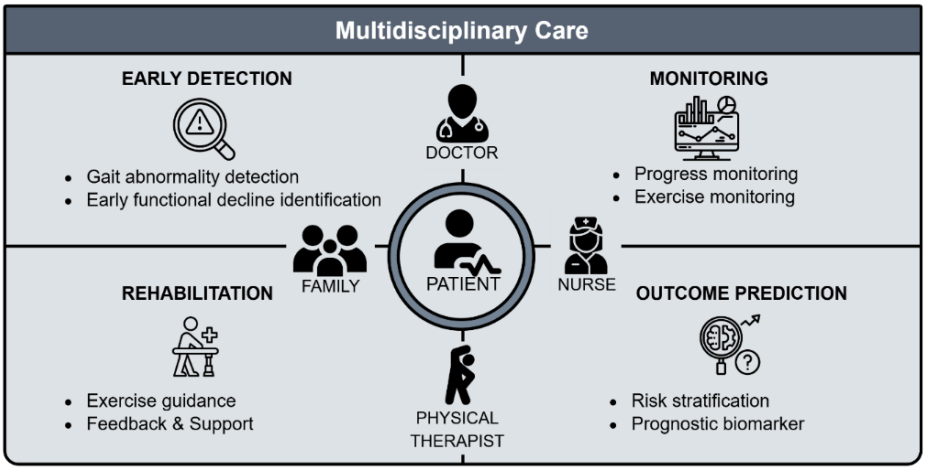

Fig. 7. Clinical evaluation and clinical expansion roadmap

3. Clinical Application & System Interoperability

- Clinical field application of high-performance insole prototypes and verification to ensure compatibility.

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	- This process involves establishing robust data synchronization between the wearable hardware and hospital, ensuring data integrity during clinical trials, collecting comprehensive feedback from diverse stakeholder groups, including physicians, nurses, physical therapists, and patients. - The roadmap includes strategic directions for applying the technology to neurodegenerative disorders like Parkinson's disease and musculoskeletal conditions such as sarcopenia, leveraging the platform's cross-disciplinary diagnostic capabilities. **4. Global Certification & Market Entry** - Obtaining official testing and certifications to facilitate product launches and exports. **Research Institute** **Enhance clinician-informed AI models and establish a foundation model with a specialized gait analysis fine-tuning framework** **1. Expert-Driven Labeling Calibration** - Introduction of a clinician-led labeling correction methodology for clinically significant gait events, leveraging specialized medical expertise. **2. Explainable AI (XAI) & Robust Feature Extraction** - Ensuring clinical reliability through the implementation of Explainable AI (XAI) and achieving robustness by enhancing the performance of universal gait feature extraction. - Furthermore, the feature extraction layer will be designed to be invariant to confounding factors like body weight or walking surface, ensuring consistent performance across diverse patient demographics. **3. Foundation Model Acquisition & Fine-Tuning** - Acquisition of a foundation model for expansion into diverse gait task domains and validation of advanced fine-tuning techniques. - By applying strategic fine-tuning methods such as LoRA (Low-Rank Adaptation) or prompt-based tuning, we aim to rapidly specialize the general model for niche clinical tasks—such as post-operative recovery monitoring or early-stage Parkinson's detection—with minimal task-specific labeled data.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**1. Design and implementation of prognostic prediction models** The institute will design and implement a predictive model that utilizes gait sensor data and clinical information to estimate postoperative functional outcomes (responder/non-responder) in patients with lower-limb joint disorders. In addition to core model development, the institute will be responsible for data preprocessing, normalization, and feature encoding of multichannel time-series sensor signals to ensure data quality and robustness. **2. Application of model compression and optimization techniques** To ensure practical deployment in clinical and commercial environments, the institute will investigate and apply various model compression techniques, including quantization, pruning, and knowledge distillation. These efforts aim to reduce model size and computational cost while maintaining predictive accuracy and clinical reliability. Furthermore, the institute will analyze trade-offs between model complexity and performance and provide guidelines for selecting appropriate lightweight model configurations under different deployment constraints. **3. Reduction of computational complexity and hardware-friendly optimization**

	The institute will focus on reducing computational complexity through efficient network architecture design, utilization of sparse matrix representations, and approximation techniques for computational efficiency. From a hardware perspective, the institute will optimize the model to ensure suitability for smart insole–based AI systems, enabling efficient on-device or edge-level inference.
Expected R&R of Korean SME Partner	**1. Clinical data acquisition and validation of clinical effectiveness** The SME partner will lead the acquisition of gait sensor data and associated clinical data using smart insoles in real-world and clinical environments. The partner will be responsible for evaluating and validating the clinical effectiveness of the developed smart insole system and AI prediction model, including usability, reliability, and consistency of measurement outcomes. In addition, the SME partner will provide structured feedback to the research institute based on deployment and testing results to support iterative model and system refinement. **2. Production and supply of smart insoles for AI research and validation** The SME partner will manufacture and supply smart insoles required for AI model development, training, and validation. This includes ensuring sensor stability, measurement accuracy, device consistency, and hardware reliability across multiple production batches. **3. Supply of data collection solutions for AI research and clinical studies, and provision of technical support** The SME partner will provide end-to-end data collection solutions to support AI research and clinical studies based on smart insole sensing systems. The SME partner will provide technical support for device operation, calibration, maintenance, and troubleshooting during experimental and clinical trials, ensuring data quality, protocol compliance, and operational reliability. **4. Development and deployment of commercialization services for integrated AI solution linkage** The SME partner will develop and deploy commercialization-oriented services that integrate the smart insole hardware with the AI-based prognostic prediction model. This includes backend system development, user interface design, and API-level linkage for real-time or batch inference services. **5. Development of high-performance smart insole prototypes and establishment of product commercialization** The SME partner will develop next-generation, high-performance smart insole prototypes based on technical feedback from model analysis and system integration. Through these activities, the SME partner will contribute to the practical translation of research outcomes into deployable healthcare products.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Clinical and Technical Utilization** **1. Quantitative gait assessment and prognostic evidence** This study enables quantitative assessment of gait function changes in lower-limb joint disorders using deep learning–based digital biomarkers derived from wearable sensor data. By supporting objective prognosis prediction and early identification of high-risk groups, the proposed approach provides clinically meaningful evidence for patient monitoring and outcome assessment across clinical and home environments. **2. Clinical decision support and advancement of digital health systems** By integrating sensor-based gait indicators with clinical variables, the proposed framework supports patient-specific rehabilitation planning and provides quantitative evidence for clinical decision-making, including the timing and methods of invasive treatments such as surgery. Furthermore, the developed technology establishes a foundation for automated gait analysis and risk prediction systems applicable to in-hospital and remote digital healthcare services. **3. Data and platform construction and utilization** The integrated gait and clinical dataset constructed through retrospective and prospective data collection will serve as a high-

	quality structured database applicable to future research topics and data-driven healthcare applications. **Policy and Industrial Utilization** **1. Utilization in healthcare policies for maintaining gait function and preventing functional decline in older adults** In the context of an increasing elderly population and a growing number of patients with gait disorders and lower-limb joint diseases, the results of this study can be used to establish prevention-oriented rehabilitation services and to formulate healthcare policies aimed at maintaining gait function in older adults. **2. Establishment of a foundation for technology commercialization** The outcomes of this study, which integrates sensor-based gait analysis and AI technologies, will provide a foundation for technology commercialization in the medical device and digital rehabilitation sectors. The developed approach enables accessible wearable healthcare solutions and evidence-based early prediction services for age-related gait disorders, contributing to early intervention and reduced medical and social costs. Furthermore, the project will support the development of clinically applicable AI-based diagnostic support tools and strengthen the competitiveness of domestic SMEs through productization and industrial collaboration. The outcomes are also expected to facilitate expansion into global healthcare markets through joint commercialization with international research partners. **Technological achievements** This project will develop a lightweight AI model for gait analysis that enables efficient and reliable inference on edge-level healthcare applications. The developed technology will have immediate application potential in the healthcare and rehabilitation markets, including hospitals, rehabilitation centers, and home-care solutions for post-surgery and neurological patients. The project is expected to contribute to healthcare cost reduction by enabling early detection of gait abnormalities and remote monitoring, thereby reducing hospital visits and long-term rehabilitation costs.

6. References

[1] Almuteb, Ibrahim, Rui Hua, and Ya Wang, "Smart insoles review (2008-2021): Applications, potentials, and future." Smart Health 25 (2022): 100301.

[2] W. Lee, et al, 'Distinguishing Pathologic Gait in Older Adults Using Instrumented Insoles and Deep Neural Networks," IEEE J Biomed Health Inform, 2025, Jul: 29(7): 4758-4768.

□ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization